

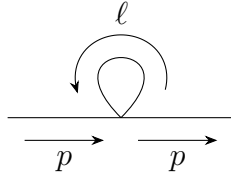
1. For $g = 0$ we have the following Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4.$$

In $d = 3$, the kinetic term gives us the dimension of the field $[\phi] = \frac{1}{2}$, therefore $[m] = 1$ and $\lambda = 1$, which means this theory is *super-renormalizable*. Using Weinberg's power counting rule, we have

$$[\Gamma_n] = 3 - \frac{1}{2}(\#\text{external lines}) - (\#\text{vertices})$$

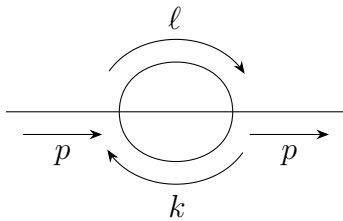
Given the interaction $\propto \phi^4$, the irreducible graphs can only have an even number of external lines. Therefore, for Γ_4 the only divergent irreducible graph is with one vertex; however, this is a three-level graph, and therefore it's convergent. The only graphs which are divergent are therefore the 2-point functions with one and two vertices:



1-loop contribution

$$[\Gamma_2^{(1)}] = 3 - \frac{1}{2}2 - 1 = 1$$

$$\Gamma_2^{(1)} = C^{(1)} + \Gamma_2^{(1)\text{reg}}$$



2-loop contribution

$$[\Gamma_2^{(2)}] = 3 - \frac{1}{2}2 - 2 = 0$$

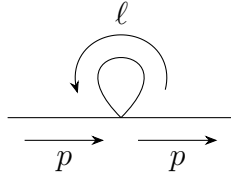
$$\Gamma_2^{(2)} = C^{(2)} + \Gamma_2^{(2)\text{reg}}$$

Overall, only the mass term Z_m has to be modified and only in the first and second order in perturbation λ , which corresponds to this theory being *super-renormalizable*.

In $d = 4$, we have $[\phi] = 1$, and therefore $m = 1$ and $\lambda = 0$, which means this theory should be *renormalizable*. Again, using Weinberg's power counting rule, we have

$$[\Gamma_n] = 4 - (\#\text{external lines})$$

From this relation, we see that all 1PI graphs with less than or equal to 4 are divergent while graphs with more than 4 external lines are convergent. Specifically, only the non-tree-level graphs with 4 external legs are divergent. In this case, the 2-point functions have the superficial degree of divergence $[\Gamma_2^{(l)}] = 2$, which causes both the mass term Z_m and the kinetic term Z_ϕ to have to be modified. These modifications start in the first order of perturbation:

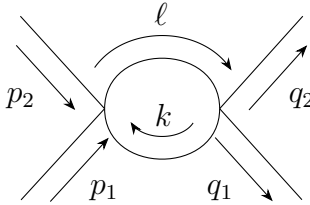


1-loop contribution

$$[\Gamma_2^{(1)}] = 4 - 2 = 2$$

$$\Gamma_2^{(1)} = C_m^{(1)} + C_\phi^{(1)} p^2 + \Gamma_2^{(1)\text{reg}}$$

Similarly, for all 1PI graphs with two external legs and more vertices, resulting in an infinite perturbation series expansion in Z_m and Z_ϕ . However, in the case of $d = 4$, the non-tree-level 1PI graphs with four external legs are also divergent, starting at one-loop:



1-loop contribution

$$[\Gamma_4^{(1)}] = 4 - 4 = 0$$

$$\Gamma_4^{(1)} = C_\lambda^{(1)} + \Gamma_4^{(1)\text{reg}}$$

In this case, the Z_λ term in the Lagrangian must be modified, again in an infinite order in perturbation.

[2.] In the case of a general g, λ in $d = 6$, we have the following Lagrangian density

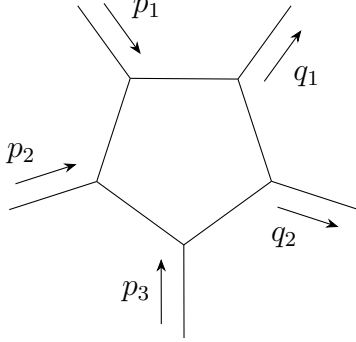
$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{g}{3!}\phi^3 - \frac{\lambda}{4!}\phi^4.$$

In $d = 6$, the kinetic term gives us $[\phi] = 2$, which results in $[m] = 1$, $[g] = 0$ and $[\lambda] = -2$, which means that this theory for $\lambda \neq 0$ should be *non-renormalizable*.

In this generic theory, 1PI graphs with both an even and odd number of external legs are allowed. In general, we have

$$[\Gamma_n] = 6 - 2(\#\text{external lines}) + 2(\#\lambda\text{-type vertices})$$

From this expression, we see that the only convergent 1PI non-tree-level graphs have to have $(3 + n_\lambda + 1)$ external legs, where n_λ is the number of λ -type vertices in the graph, for example:



$$[\Gamma_5^{(1)}] = 6 - 2 \cdot 5 + 2 \cdot 0 = -4$$

$$\Gamma_5^{(1)} = \Gamma_5^{(1)\text{reg}}$$

However, for each n -point function, we can find a 1PI graph with a high enough number of λ -vertices such that the n -point function is divergent. Therefore, all operators, even the ones now written with zero coefficients in the Lagrangian density, will need modification in infinite order in perturbation.

[3.] In $d = 3$ the kinetic term gives us $[\phi] = \frac{1}{2}$, $[m] = 1$. For the Lagrangian to respect the global $\mathbb{Z}_2$ symmetry, we can only have terms which are even in the powers of the field ϕ . In addition, for the theory to be renormalizable, the operators in the Lagrangian can have at most order $[\mathcal{O}] = d = 3$. This results in only two interaction operators ϕ^4 and ϕ^6 . In terms of power counting and $\mathbb{Z}_2$ symmetry, the term $(\partial\phi)\phi^3$ could also come to mind; however, these terms could be written as a total four-divergence $\frac{1}{4}\partial(\phi^4)$ and therefore would not change the action. Overall, we obtain the following Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 - \frac{\kappa}{6!}\phi^6 - \Omega_0.$$

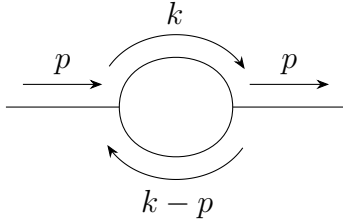
[4.] For $\lambda = 0$ the Lagrangian has the following form

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{g}{3!}\phi^3.$$

In $d = 6$ we have $[\phi] = 2$, $[m] = 1$ and $[g] = 0$, and therefore this theory should be *renormalizable*. Specifically, the Weinberg's power-counting gives us

$$[\Gamma_n] = 6 - 2(\#\text{external lines}),$$

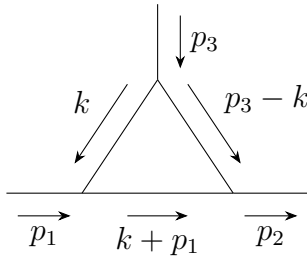
which results in Γ_2 being quadratically divergent and Γ_3 being logarithmically divergent. Specifically, for one-loop 1PI graphs, we have:



1-loop contribution

$$[\Gamma_2^{(1)}] = 6 - 2 \cdot 2 = 2$$

$$\Gamma_2^{(1)} = C_m^{(1)} + C_\phi^{(1)} p^2 + \Gamma_2^{(1)\text{reg}}$$



1-loop contribution

$$[\Gamma_3^{(1)}] = 6 - 2 \cdot 3 = 0$$

$$\Gamma_3^{(1)} = C_g^{(1)} + \Gamma_3^{(1)\text{reg}}$$

Overall, we see that for the 2-point function, the coefficients Z_m and Z_ϕ will have to be modified, and for the 3-point function, we will have to modify the Z_g coefficient.

Firstly, let us isolate the divergent term in the 2-point function. Let us denote

$$i\Gamma_2^{(1)}(p) = i(p^2 - m^2 + \Sigma(p))$$

and now we compute

$$\begin{aligned} i\Sigma_2(p) &= \frac{1}{2}(-ig)^2 \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2 + i0} \frac{i}{(p-k)^2 + m^2 + i0} \\ &= \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 - m^2 + i0} \frac{1}{(p-k)^2 + m^2 + i0} \\ &= \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \int_0^1 dx \int \frac{d^d k}{(2\pi)^d} \frac{1}{[k^2 + x(1-x)p^2 - m^2]^2} \\ &= \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \int_0^1 dx \int \frac{d^d k_E}{(2\pi)^d} \frac{1}{[k_E^2 - (x(1-x)p^2 - m^2)]^2} \\ &= i \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \int_0^1 dx \int \frac{d^d k_E}{(2\pi)^d} \frac{1}{[k_E^2 + \Delta]^2} \end{aligned}$$

$$\begin{aligned} &= i \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \int_0^1 dx \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(2-d/2)}{\Gamma(2)} \Delta^{d/2-2} \\ &= i \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \frac{1}{(4\pi)^{3-\epsilon}} \Gamma(\epsilon-1) \int_0^1 dx \Delta^{1-\epsilon} \end{aligned}$$

Now we have to identify the divergent term, which clearly stems from the $\Gamma(\epsilon-1)$, as we have

$$\begin{aligned} \mu^{2\epsilon} &\approx 1 + 2\epsilon \ln \mu + \mathcal{O}(\epsilon^2), \\ (4\pi)^{-\epsilon} &\approx 1 - \epsilon \ln 4\pi + \mathcal{O}(\epsilon^2), \\ \Gamma(\epsilon-1) &= \frac{\Gamma(\epsilon)}{\epsilon-1} \approx -\frac{1}{\epsilon} + \gamma_E + \mathcal{O}(\epsilon), \\ \Delta^{-\epsilon} &\approx 1 - \epsilon \ln \Delta + \mathcal{O}(\epsilon^2). \end{aligned}$$

Therefore, we obtain

$$i\Sigma_2^{\text{div}} = -i \frac{\hat{g}^2}{2(4\pi)^3} \frac{1}{\epsilon} \int_0^1 dx [-x(1-x)p^2 + m^2] = -i \frac{\hat{g}^2}{2(4\pi)^3} \frac{1}{\epsilon} \left[m^2 - \frac{p^2}{6} \right].$$

Using the minimal subtraction renormalization, we finally obtain

$$\begin{aligned} Z_\phi &= 1 - \frac{\hat{g}^2}{12(4\pi)^3} \frac{1}{\epsilon}, \\ Z_m &= 1 + \frac{\hat{g}^2}{2(4\pi)^3} \frac{1}{\epsilon}. \end{aligned}$$

We can use a similar procedure for the renormalization of the 3-point function. Here we denote

$$i\Gamma_3^{(1)}(p_1, p_3) = i(g + \Sigma(p_1, p_3))$$

and compute

$$\begin{aligned} i\Sigma &= (-ig)^3 \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2} \frac{i}{(k+p_1)^2 - m^2} \frac{i}{(k-p_3)^2 - m^2} \\ &= g^3 \mu^{3\epsilon} \Gamma(3) \int_0^1 dx \int_0^1 dy \int_0^1 dz \int \frac{d^d k}{(2\pi)^d} \delta(1-x-y-z) \frac{1}{(\dots)^3} \\ &= 2ig^3 \mu^{3\epsilon} \int_0^1 dx \int_0^{1-x} dy \int \frac{d^d k_E}{(2\pi)^d} \frac{1}{[k_E^2 + \Delta]^3}, \end{aligned}$$

where we have defined

$$\Delta = m^2 - [y(1-y)p_1^2 + (1-x-y)(x+y)p_3^2 + 2y(1-x-y)p_1 \cdot p_3].$$

Now, we have

$$\begin{aligned} i\Sigma &= 2ig^3\mu^{3\epsilon}\frac{1}{(4\pi)^{d/2}}\frac{\Gamma(3-d/2)}{\Gamma(3)}\int_0^1 dx \int_0^{1-x} dy \Delta^{d/2-3} \\ &= ig^3\mu^{3\epsilon}\frac{1}{(4\pi)^{3-\epsilon}}\Gamma(\epsilon)\int_0^1 dx \int_0^{1-x} dy \Delta^\epsilon. \end{aligned}$$

Here again using the MS, we extract only the divergent term. It is worth noting that due to the logarithmic character of the divergence, the divergent term does not depend on the specific form of $\Delta(p_1, p_3, m)$, and therefore the precise computation of the Feynman parametrization could have been omitted.

$$i\Sigma_3^{\text{div}} = g^3 \frac{1}{(4\pi)^3} \frac{1}{\epsilon} \int_0^1 dx \int_0^{1-x} dy 1 = \frac{g^3}{2(4\pi)^3} \frac{1}{\epsilon},$$

which gives the following renormalization term

$$Z_g = g - \frac{g^3}{2(4\pi)^3} \frac{1}{\epsilon}.$$